

Capstone Expo

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Improve and Challenge

Without opening the old lesson, explain one key idea from Improve and Challenge. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Build the Recommender

Without opening the old lesson, explain one key idea from Build the Recommender. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Agents and Tools

Without opening the old lesson, explain one key idea from Agents and Tools. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Check report consistency for Defending the data application. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Defending the data application. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Defend a new case independently

Limit is 10. A has minutes 10, fun 2, learning 4; B has 9,3,2; C has 11,5,5; D has 8,"4",2. Validate, rank, give counts, explain the winner, then reverse the input order. Present without hints.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Present a recommender that gets 8/10 final requests right. One failure invents a title; another violates a hard page limit. Is '80%, ready for everyone' justified? Propose release criteria.

6. Solve the given case

An expo claim says the recommender knows everybody's favourite book after twelve tests.

Rewrite the defence.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Your demo recommends the right book once. An audience member asks why they should trust it. Name evidence and one limitation you should present.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 36 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Final evidence bundle: brief, data, scoring logic, tests, improvement record, and product card.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Present a recommender that gets 8/10 final requests right. One failure invents a title; another violates a hard page limit. Is '80%, ready for everyone' justified? Propose release criteria.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Assemble evidence

List the files or records a peer needs to reproduce the chooser result.

My reasoning and evidence

13. W2 Check report consistency

A report says supplied 5, accepted 4 and rejected 2. Identify the inconsistency.

My reasoning and evidence

14. W3 Repair a claim

A presentation says the project proves students learn better. What claim does a successful rule-based demo support instead?

My reasoning and evidence

15. W4 Defend a new case independently

Limit is 10. A has minutes 10, fun 2, learning 4; B has 9,3,2; C has 11,5,5; D has 8,"4",2. Validate, rank, give counts, explain the winner, then reverse the input order. Present without hints.

My reasoning and evidence

16. W5 State the next evidence need

After passing the new case, what remains unproven and what is a reasonable next step?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Capstone Expo

A fictional team presents its transparent book recommender on a reserved catalogue.

Request maximum 120 pages; $\text{score} = 2 \times \text{interest} + \text{illustrated}$, where illustrated is 1 for True and 0 for False; tie by ID.

A pages120 interest3 True; B pages100 interest4 False; C pages121 interest5 True; D pages90 interest"4" True. Numeric fields require exact integers.

Capstone Expo

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Validate

D has text interest and is rejected; A,B,C have valid fields.



2. Filter

A and B meet the page limit; C exceeds it.



3. Rank

A scores 7; B scores 8, so B wins.



4. Defend evidence

Explain two eligible items, one page-limit exclusion and one type rejection.

CHANGE ONE THING

An authorised repair converts D's interest to integer 4.

Explain the changed result using the matching lesson.

Give the original winning ID, its score, and counts of eligible, excluded and invalid rows.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Present the Transparent Book Recommender's code, baseline, full-case traces, reserved results and limitations, without claiming fictional results as pupil work.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

Present the Transparent Book Recommender's code, baseline, full-case traces, reserved results and limitations, without claiming fictional results as pupil work.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional team presents its transparent book recommender on a reserved catalogue.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Explain two eligible items, one page-limit exclusion and one type rejection.

1. Defend evidence
2. Validate
3. Filter
4. Rank

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Defend evidence Explain two eligible items, one page-limit exclusion and one type rejection. A presentation should let someone rebuild the result from the stated contract. A's equality is accepted; D is not silently converted; C's high preference score cannot override the page requirement. Reversing rows must preserve B because it has the unique highest eligible score. Keep the reserved case distinct from practice demonstrations.

Practice 2: explain the deciding evidence

Changed-case practice: An authorised repair converts D's interest to integer 4. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. D becomes eligible and scores 9, so it wins.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: D becomes eligible and scores 9, so it wins. The evidence-backed input repair changes validation and then ranking; record this as a new case/version.

Repair a missing building block

Review Improve and Challenge (ai-2/35). Goal: Repair a specific observed failure. Short practice: Why are tests at both 100 and 101 needed?. Return to this week and explain what you changed.

Review Build the Recommender (ai-2/33). Goal: Implement a deterministic recommender. Short practice: Explain why B does not win after D is excluded even though it has fewer pages.. Return to this week and explain what you changed.

Review Agents and Tools (ai-2/30). Goal: Explain the mechanism and terminology of state machines for bounded agents. Short practice: Can the assistant accept Trace B because its final state is done?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
